



OpsSphere

Software Requirements Specification

Enterprise Support Operations Platform

Formal consolidated requirements artifact based on business discovery, project definition, requirements analysis, use cases, business rules, scope, and success criteria.

Field	Value
Project	OpsSphere
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Business Context	Multinational BPO / Contact Center Operations

Prepared for portfolio presentation and implementation reference.

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1. Document Control

1.1 Document Information

Field	Value
Project Name	OpsSphere - Enterprise Support Operations Platform
Project Type	Senior-level full stack enterprise software project
Document Owner	Technical Owner
Status	Draft
Version	1.0
Primary Audience	Portfolio reviewers, technical interviewers, implementation agents, and project stakeholders
Repository Target	docs/formal-specification/

1.2 Version History

Version	Date	Description	Author
1.0	2026-05-13	Initial consolidated formal SRS draft for OpsSphere.	Technical Owner

1.3 Intended Audience

- Technical reviewers evaluating requirements quality, system thinking, and implementation readiness.
- Portfolio reviewers evaluating business analysis, enterprise architecture, and senior-level full stack planning.
- Implementation agents that need a consolidated reference before coding.
- Future maintainers who need to understand scope, rules, and acceptance expectations.

1.4 Source Documents

Source Document	Contribution to This SRS
docs/00-executive-summary.md	Project overview, business value, portfolio purpose, core capabilities, and technical vision.
docs/01-business-context.md	Operating model, organizational hierarchy, business domain, and reporting boundary.
docs/02-business-case.md	Business problem, proposed solution, value drivers, risks reduced, and measurable operational improvements.
docs/03-project-charter.md	Project goals, general scope, assumptions, constraints, risks, and success criteria.
docs/04-stakeholders.md	Stakeholder roles, needs, pains, responsibilities, and interaction expectations.
docs/05-scope-and-roadmap.md	MVP scope, out-of-scope boundaries, assumptions, roadmap, and release milestones.
docs/06-requirements.md	Primary requirements baseline for functional, non-functional, business rule, acceptance, and glossary content.
docs/07-use-cases.md	Use case summary, actor goals, flows, business rules, and acceptance criteria.
docs/09-business-rules.md	Operational constraints, validation rules, authorization rules, workflow rules, and audit rules.

1.5 Endpoint Convention Note

This SRS resolves the documentation inconsistency found during audit regarding the health endpoint path. Operational health endpoints are treated as technical readiness/liveness endpoints and should not be modeled as business API resources.

Endpoint	Purpose	Decision
GET /health	Readiness/liveness health check for API process, database connectivity, and required configuration.	Canonical endpoint for MVP.
GET /health/details	Detailed diagnostic health endpoint for development, staging, or protected operational use.	Canonical detailed endpoint.
GET /api/health	Alternative API-base-path style endpoint.	Not recommended as the default; should be avoided unless explicitly reintroduced.

2. Introduction

2.1 Project Overview

OpsSphere is an enterprise-grade operations platform designed for multinational BPO and contact center environments that manage support operations across regions, countries, client accounts, campaigns, supervisors, agents, viewers, customers, tickets, SLAs, audit history, dashboards, and role-based access control.

The platform centralizes the operational structure and ticket lifecycle of support operations. It is intended to improve ownership visibility, SLA awareness, internal collaboration, escalation handling, auditability, and reporting-ready data quality.

OpsSphere is not intended to be a generic ticketing system. It is designed as an enterprise operations platform that connects business workflows, organizational structure, accountability, auditability, and modern software architecture practices.

2.2 Purpose of the SRS

The purpose of this Software Requirements Specification is to define what OpsSphere must do, what quality conditions it must satisfy, what business rules it must enforce, and what acceptance criteria should be used to validate the initial version.

This document serves as a consolidated formal reference for implementation, testing, architectural review, and portfolio presentation.

2.3 Product Vision

The product vision is to provide a maintainable, secure, testable, and enterprise-ready platform that allows a multinational BPO/contact center organization to manage support execution across its operational hierarchy.

2.4 Document Scope

This SRS covers the initial MVP requirements for OpsSphere, including authentication, authorization, user management, organizational structure management, customer management, ticket lifecycle management, SLA tracking, internal comments, escalations, audit history, operational dashboards, filtering, and reporting-ready data.

This SRS does not define final implementation code, final database migrations, final UI wireframes, or production operational runbooks. Those items are covered by other design and delivery documents.

2.5 Portfolio Purpose

OpsSphere is also a portfolio artifact. It demonstrates the ability to translate a realistic business problem into a structured software solution using business analysis, requirements engineering, process modeling, domain modeling, Clean Architecture, role-based authorization, automated testing, DevOps, and Azure-ready planning.

3. Business Context

3.1 Business Background

OpsSphere is modeled around a multinational BPO/contact center organization that provides operational support services for multiple client accounts across different regions, countries, campaigns, supervisors, and agents.

The business operates through an internal support structure where agents handle tickets, supervisors oversee operational execution, operations managers review performance across regions, and administrators maintain the organizational configuration of the platform.

3.2 Business Problem

- Operational data may be fragmented across disconnected tools, spreadsheets, email threads, chat messages, or incomplete reporting sources.
- Ticket ownership, account responsibility, campaign workload, supervisor accountability, and SLA state may be difficult to determine.
- Manual reporting dependencies can create inconsistent operational visibility and unreliable analytics.
- Auditability may be weak when ownership changes, comments, status transitions, and escalations are not recorded consistently.
- Administrative structure can become outdated when regions, countries, accounts, campaigns, roles, and assignments are not centrally maintained.

3.3 Proposed Solution

OpsSphere provides a centralized operations platform that captures the operational hierarchy, controls user access, supports ticket execution, tracks SLA state, preserves audit history, and produces structured data that can later support reporting and analytics tools such as Power BI.

3.4 Business Value

- Centralizes ticket, customer, account, campaign, and user management.
- Improves ticket ownership visibility across regions, countries, accounts, campaigns, supervisors, and agents.
- Supports SLA monitoring for tickets that are within SLA, at risk, breached, or completed.
- Strengthens accountability through audit logging for important business actions.
- Improves supervisor and manager oversight through dashboards and filtered views.
- Provides cleaner operational data for external reporting tools.

3.5 Business Metrics Supported

Metric Area	Example Metrics
Ticket Volume	Open tickets, created tickets, resolved tickets, closed tickets.
SLA Performance	Within SLA, at risk, breached, completed within SLA, completed after breach.
Workload	Tickets by agent, supervisor, account, campaign, priority, and status.
Operational Risk	Overdue tickets, escalated tickets, aging tickets, overloaded teams.
Auditability	Critical actions by actor, timestamp, entity, previous value, and new value.

4. Project Scope

4.1 In Scope

- Authentication and protected application access.
- Role-based access control and scope-based visibility.
- User, role, permission, and operational scope management.
- Region, country, account, campaign, manager, supervisor, agent, and viewer management.
- Customer management to support ticket context.
- Ticket lifecycle management from creation to closure.
- Ticket assignment, status workflow, priority, comments, escalation, resolution, and closure.
- Basic SLA target and SLA state tracking.
- Audit history for critical operational and administrative changes.
- Basic operational dashboards, filtering, and reporting-ready data.

4.2 Out of Scope

- Full business intelligence platform or advanced Power BI dashboard creation.
- Payroll, HR, workforce management scheduling, call recording, telephony, and omnichannel messaging integrations.
- External customer portal, mobile application, real-time chat, and production-grade enterprise SSO.
- AI ticket classification, predictive SLA modeling, advanced automation workflows, and multi-tenant billing model.
- Replacement for CRM systems, Power BI, workforce management tools, or full analytics platforms.

4.3 MVP Scope

The MVP must prove that OpsSphere can represent a realistic BPO/contact center hierarchy and support the core ticket lifecycle with secure access, operational visibility, SLA tracking, comments, escalations, and audit history.

4.4 Future Roadmap Summary

Phase	Summary
MVP	Core operational foundation: authentication, RBAC, structure management, customers, tickets, SLA, comments, audit, dashboards.
Phase 2	Enhanced dashboards, escalation queues, notification center, attachments, improved supervisor workflows, and export quality.
Phase 3	AI-assisted classification, predictive SLA risk, selected integrations, Azure deployment pipeline, and production hardening.
Future Considerations	Customer portal, mobile app, knowledge base, RAG assistant, advanced reporting, QA workflows, and integrations.

4.5 Scope Control Principles

- Build the operational foundation before advanced analytics or automation.
- Prioritize clear workflows over visual complexity.
- Use fictional sample data only.
- Design enterprise-ready architecture but deliver incrementally.
- Keep Power BI and advanced executive reporting outside the core MVP.

5. Stakeholder Overview

5.1 Stakeholder Summary

Stakeholder	Primary Responsibility	Access Type
Admin	Configure and maintain the platform structure, users, roles, permissions, and operational catalogs.	Global administrative read/write.
Operations Manager	Monitor performance across assigned regions and review workload, SLA, escalations, and reporting views.	Read and oversight within assigned regions.
Supervisor	Manage agents and tickets within assigned accounts or campaigns.	Operational read/write within scope.
Agent	Handle tickets within assigned account or campaign.	Limited write within assigned scope.
Viewer	Review dashboards, tickets, reports, and audit history without modifying records.	Read-only within assigned scope.
Customer	Associated with support cases, but does not directly access the MVP.	No direct runtime access.
Technical Owner	Design, build, test, maintain, and evolve the software solution.	Implementation role, not a business runtime role.

5.2 Stakeholder Needs

Stakeholder	Key Needs
Admin	Reliable structure management, validation, permissions, role assignment, user activation/deactivation, and audit history.
Operations Manager	Regional visibility, filters, SLA trends, workload review, escalation visibility, and reporting-ready data.
Supervisor	Agent workload visibility, assignment tools, escalation review, SLA risk visibility, and audit trail access.
Agent	Clear assigned ticket queue, status updates, comments, SLA targets, escalation path, and simple ticket workflows.
Viewer	Read-only visibility into operational data, dashboards, reports, and audit history within assigned scope.

5.3 Access and Visibility Summary

All runtime users must authenticate before accessing protected features. Runtime access must be determined by role, permission, and operational scope. Frontend visibility may improve user experience, but backend authorization remains the security boundary.

6. Product Perspective

6.1 System Context

OpsSphere consists of an Angular frontend, an ASP.NET Core Web API backend, application use cases, domain logic, Entity Framework Core persistence, SQL Server storage, authentication and authorization services, audit logging, operational dashboards, and reporting-ready data views.

[Graphic placeholder: C4 system context diagram can be inserted here.]

6.2 Operational Hierarchy

The operational hierarchy is represented as:

Multinational BPO / Contact Center -> Region -> Country -> Account -> Campaign -> Supervisor -> Agent

[Graphic placeholder: operational hierarchy diagram can be inserted here.]

6.3 Internal Users

The initial version is for internal operational users only: Admin, Operations Manager, Supervisor, Agent, and Viewer. Customers are associated with tickets but do not directly access the platform in the MVP.

6.4 External Systems and Boundaries

External System	MVP Relationship
SQL Server	Primary relational database for operational data.
Browser	User access to Angular frontend.
Power BI	External reporting and advanced analytics boundary, not replaced by OpsSphere.
Azure App Service / Azure SQL	Future Azure-ready deployment target.
Azure Key Vault	Future secrets management target.
Application Insights	Future telemetry target.
Enterprise Identity Provider	Future SSO integration, not MVP.

6.5 Operational Health Endpoints

Health checks are operational endpoints used for deployment validation, readiness, liveness, and support diagnostics. They should be exposed outside the business API resource model.

Endpoint	Requirement
GET /health	Returns health status for the API process, database connectivity, and required configuration.
GET /health/details	Returns detailed health information when enabled and protected appropriately.

7. Product Features Summary

Feature	Summary
Authentication and Authorization	Secure login, JWT authentication, protected endpoints, role-based access, and scope-based visibility.
User and Role Management	Admin-managed users, roles, permissions, activation, deactivation, and scope assignment.
Organizational Structure Management	Regions, countries, accounts, campaigns, managers, supervisors, agents, viewers, roles, and permissions.
Customer Management	Customer creation, update, ticket linkage, and scoped customer history.
Ticket Lifecycle Management	Create, assign, update status, prioritize, comment, escalate, resolve, close, and track history.
SLA Tracking	SLA target assignment, state tracking, at-risk visibility, breached visibility, and final outcome preservation.
Internal Comments	Internal collaboration notes tied to ticket, author, timestamp, and audit context.
Escalation Visibility	Escalation reason capture, supervisor visibility, escalation queue, and audit trail.
Audit History	Traceability of critical ticket, user, role, scope, permission, and structure changes.
Operational Dashboards	Open tickets, overdue tickets, ticket distribution, SLA overview, workload, and filtered views.
Reporting-Ready Data	Structured operational fields and exports that support external reporting tools.

8. Functional Requirements

8.1 Authentication and Authorization Requirements

- FR-AUTH-001: The system shall allow registered users to authenticate using valid credentials.
- FR-AUTH-002: The system shall use JWT-based authentication for API access.
- FR-AUTH-003: The system shall prevent unauthenticated users from accessing protected application features.
- FR-AUTH-004: The system shall enforce role-based access control for Admin, Operations Manager, Supervisor, Agent, and Viewer roles.
- FR-AUTH-005: The system shall restrict user visibility based on assigned operational scope.
- FR-AUTH-006: The system shall return appropriate authorization errors when users attempt to access resources outside role or scope.
- FR-AUTH-007: The system shall expire user sessions according to configured token expiration rules.

8.2 User and Role Management Requirements

- FR-USER-001: The system shall allow Admin users to create, view, update, deactivate, and manage users.
- FR-USER-002: The system shall allow Admin users to assign roles to users.
- FR-USER-003: The system shall allow Admin users to assign operational scope to Managers, Supervisors, Agents, and Viewers.
- FR-USER-004: The system shall prevent deactivated users from authenticating.
- FR-USER-005: The system shall preserve historical tickets, comments, assignments, and audit records when users are deactivated.
- FR-USER-006: The system shall record important user, role, and scope management changes in audit history.

8.3 Operational Structure Requirements

- FR-ORG-001: The system shall allow Admin users to create, view, update, and deactivate regions.
- FR-ORG-002: The system shall allow Admin users to create, view, update, and deactivate countries.
- FR-ORG-003: The system shall allow Admin users to assign countries to regions.
- FR-ORG-004: The system shall allow Admin users to create, view, update, and deactivate accounts.
- FR-ORG-005: The system shall allow Admin users to create, view, update, and deactivate campaigns.
- FR-ORG-006: The system shall allow Admin users to assign managers, supervisors, agents, and viewers to operational scopes.
- FR-ORG-007: The system shall preserve historical context when organizational records are deactivated.
- FR-ORG-008: The system shall record important organizational structure changes in audit history.

8.4 Customer Management Requirements

- FR-CUST-001: The system shall allow authorized users to create customer records.
- FR-CUST-002: The system shall allow authorized users to view customer records within assigned scope.
- FR-CUST-003: The system shall allow authorized users to update customer records within assigned scope.
- FR-CUST-004: The system shall link tickets to customers.
- FR-CUST-005: The system shall allow authorized users to view customer ticket history within assigned scope.
- FR-CUST-006: The system shall prevent customers from directly accessing the MVP.

8.5 Ticket Management Requirements

- FR-TICKET-001: The system shall allow authorized users to create tickets.
- FR-TICKET-002: The system shall require customer, account, campaign, category, priority, description, and creator information when creating a ticket.
- FR-TICKET-003: The system shall assign or preserve ticket context for region, country, account, campaign, customer, supervisor, and assigned agent when applicable.
- FR-TICKET-004: The system shall allow authorized users to view tickets within assigned role and scope.

- FR-TICKET-005: The system shall allow authorized users to update editable ticket fields according to role, permission, scope, and status.
- FR-TICKET-006: The system shall allow supervisors to assign and reassign tickets to eligible active agents within scope.
- FR-TICKET-007: The system shall prevent assignment to agents outside the correct account or campaign scope.
- FR-TICKET-008: The system shall allow authorized users to change ticket status according to valid workflow transitions.
- FR-TICKET-009: The system shall prevent closed tickets from being modified unless reopening is explicitly supported and authorized.
- FR-TICKET-010: The system shall preserve ticket history for assignments, status changes, priority changes, escalations, resolutions, and closure.

8.6 Ticket Status Workflow Requirements

Status	Description
Open	Ticket has been created and awaits assignment or work.
Assigned	Ticket has an assigned agent.
In Progress	Assigned user is actively working the ticket.
Waiting for Customer	Ticket requires additional customer information or response.
Escalated	Ticket requires supervisor or higher-level attention.
Resolved	Issue has been handled and awaits closure.
Closed	Ticket lifecycle is complete and normal edits are blocked.

The system shall validate workflow transitions and reject invalid status changes. A ticket must be resolved before it can be closed unless a future authorized exception is explicitly designed.

8.7 Internal Comment Requirements

- FR-COMMENT-001: The system shall allow authorized users to add internal comments to tickets within scope.
- FR-COMMENT-002: The system shall require comment content to be non-empty.
- FR-COMMENT-003: The system shall capture comment author and timestamp.
- FR-COMMENT-004: The system shall prevent Viewers from adding comments.
- FR-COMMENT-005: The system shall record important comment events in audit history when required.

8.8 Escalation Requirements

- FR-ESC-001: The system shall allow authorized users to escalate active tickets within scope.
- FR-ESC-002: The system shall require an escalation reason.
- FR-ESC-003: The system shall prevent closed tickets from being escalated.
- FR-ESC-004: The system shall make escalated tickets visible in supervisor escalation views.
- FR-ESC-005: The system shall record escalation events in ticket history and audit history.

8.9 Resolution and Closure Requirements

- FR-RES-001: The system shall allow authorized users to resolve tickets within role and scope.
- FR-RES-002: The system shall preserve resolution timestamp, resolver, and final SLA outcome.
- FR-CLOSE-001: The system shall allow authorized users to close resolved tickets.
- FR-CLOSE-002: The system shall prevent unresolved tickets from being closed.
- FR-CLOSE-003: The system shall record resolution and closure in ticket history and audit history.

8.10 SLA Requirements

- FR-SLA-001: The system shall assign an SLA target to each ticket.
- FR-SLA-002: The system shall calculate or store SLA state as Within SLA, At Risk, Breached, or Completed.
- FR-SLA-003: The system shall make SLA state visible to authorized users in ticket views, queues, and dashboards.
- FR-SLA-004: The system shall identify tickets close to SLA breach.

- FR-SLA-005: The system shall identify tickets that have breached SLA.
- FR-SLA-006: The system shall preserve final SLA outcome after resolution or closure.
- FR-SLA-007: The system shall keep MVP SLA behavior simple and defer advanced business-hour calendars and pause rules.

8.11 Dashboard and Filtering Requirements

- FR-DASH-001: The system shall provide dashboards for operational visibility.
- FR-DASH-002: Dashboards shall display open tickets, overdue tickets, tickets by status, priority, account, campaign, agent, supervisor, and SLA state when permitted by role and scope.
- FR-DASH-003: The system shall support filters for region, country, account, campaign, supervisor, agent, status, priority, SLA state, and date created.
- FR-DASH-004: Dashboard data shall respect role and operational scope restrictions.
- FR-DASH-005: Viewers shall access dashboards in read-only mode.

8.12 Audit History Requirements

- FR-AUDIT-001: The system shall record important business actions in audit history.
- FR-AUDIT-002: Audit events shall identify actor, timestamp, entity type, entity identifier, action, previous value, and new value when applicable.
- FR-AUDIT-003: Ticket creation, assignment, reassignment, status change, priority change, escalation, resolution, closure, and comments shall be auditable.
- FR-AUDIT-004: User creation, role assignment, scope assignment, deactivation, and organizational structure changes shall be auditable.
- FR-AUDIT-005: Audit records shall not be editable through normal business workflows.
- FR-AUDIT-006: Audit visibility shall respect role and operational scope.

8.13 Reporting Data Requirements

- FR-REP-001: The system shall store structured operational data suitable for reporting and analysis.
- FR-REP-002: Ticket records shall maintain consistent reporting fields such as region, country, account, campaign, supervisor, agent, customer, status, priority, SLA state, and relevant timestamps.
- FR-REP-003: The system may provide basic CSV export for ticket lists, SLA views, or audit views.
- FR-REP-004: Reporting views shall respect role and operational scope restrictions.
- FR-REP-005: OpsSphere shall not attempt to replace Power BI or become a full BI platform in the MVP.

9. Non-Functional Requirements

Category	Requirement Summary
Security	Protected features require authentication. Backend authorization must enforce role, permission, and scope. Passwords must be stored securely. Sensitive data must not be logged.
Authorization	Backend authorization is the source of truth. Frontend role-aware visibility is not a security boundary.
Auditability	Critical business and administrative actions must be traceable through immutable audit records.
Maintainability	Backend shall follow Clean Architecture. Controllers remain thin. Business rules live in domain/application layers.
Testability	Core business rules, application handlers, API behavior, authorization, persistence, and critical journeys should be covered by automated tests.
Reliability	The system should handle validation, authorization, and unexpected errors consistently without exposing sensitive internals.
Usability	Operational workflows should be clear, role-aware, and simple enough for internal users.
Performance Expectations	Dashboard, ticket list, SLA, and reporting queries should support indexed filtering and practical MVP usage.
Observability	System shall support structured logging, correlation IDs, safe error responses, health checks, audit logs, and Application Insights readiness.
Deployment Readiness	System should be Docker-ready, GitHub Actions-ready, and Azure-ready.
Data Integrity	Core relationships should preserve relational integrity, historical traceability, and deactivation instead of destructive deletion where needed.

9.1 Health and Support Requirements

- NFR-OPS-001: The system shall expose GET /health for readiness and liveness validation.
- NFR-OPS-002: The system may expose GET /health/details for detailed diagnostics when appropriately protected or disabled in production.
- NFR-OPS-003: Health checks shall verify API process availability, database connectivity, and required configuration.
- NFR-OPS-004: Error responses should include safe messages and correlation IDs when useful.
- NFR-OPS-005: Logs must not include passwords, password hashes, JWT tokens, secrets, full connection strings, or unnecessary personal data.

10. Business Rules

10.1 General Platform Rules

- All protected OpsSphere features must require user authentication.
- The initial version is intended for internal operational users only.
- Customers must not directly access OpsSphere in the MVP.
- A user must have an assigned role and operational scope before accessing protected business data.
- Users must receive only the minimum permissions required for their responsibilities.
- Operational records must be connected to the correct business context whenever applicable.
- The MVP must prioritize clear operational workflows over advanced automation or complex integrations.

10.2 Access and Authorization Rules

- Only active users may authenticate.
- Deactivated users must not authenticate or access protected features.
- Protected API endpoints must require valid authentication tokens.
- Authentication must be verified before authorization rules are applied.
- Every internal user must have at least one valid role before accessing business modules.
- Backend services must enforce authorization independently from frontend visibility.
- Users must not access records outside their assigned operational scope.

10.3 Organizational Structure Rules

- A country must belong to one region.
- An account may contain one or more campaigns.
- A campaign must belong to one account and be associated with an operating country.
- A supervisor must be assigned to an account or campaign before managing tickets for that scope.
- An agent must be assigned to an account or campaign before handling tickets.
- Only Admin users may create, update, deactivate, or assign structural records.
- Operational records should be deactivated instead of physically deleted when historical records depend on them.

10.4 Ticket and Workflow Rules

- A ticket must be linked to a customer, account, and campaign.
- A ticket must have category, priority, description, status, and creator information.
- New tickets must start as Open unless assignment rules set them directly to Assigned.
- Ticket status changes must follow valid workflow transitions.
- A ticket must be resolved before it can be closed.
- Closed tickets cannot be edited unless reopened by an authorized role.
- Ticket changes that affect business state must be recorded in ticket history and audit history.

10.5 Assignment Rules

- A ticket can only be assigned to an eligible active agent.
- The target agent must belong to the correct account or campaign scope.
- Supervisors can only assign tickets within their assigned operational scope.
- Closed tickets cannot be assigned unless a reopening workflow is explicitly allowed.
- Assignment and reassignment must be recorded in audit history.

10.6 SLA Rules

- Each ticket must have an SLA target.
- SLA timers start when the ticket is created.

- SLA state must be visible to authorized users.
- Tickets approaching breach should be marked as At Risk.
- Tickets exceeding SLA target must be marked as Breached.
- Completed tickets should preserve their final SLA outcome.
- SLA dashboard data must respect role and operational scope.

10.7 Escalation and Comment Rules

- Escalated tickets must include an escalation reason.
- Closed tickets cannot be escalated.
- Escalated tickets must appear in supervisor escalation views.
- Internal comments cannot be empty.
- Viewers must not add comments or modify operational records.
- Escalation and comment activity must be traceable when required.

10.8 Audit and Reporting Rules

- Audit records must identify actor, timestamp, entity, action, and previous/new values when applicable.
- Audit records should not be editable through normal workflows.
- Dashboards and reporting views must respect role and operational scope.
- The system must store structured operational data suitable for reporting and analysis.
- OpsSphere dashboards must provide basic operational visibility and must not replace Power BI in the MVP.

11. Use Case Summary

Use Case ID	Name	Primary Actor(s)	Goal
UC-001	Authenticate User	Admin, Manager, Supervisor, Agent, Viewer	Allow a registered internal user to securely access OpsSphere.
UC-002	Create Ticket	Agent, Supervisor	Create a ticket linked to customer and operational context.
UC-003	Assign Ticket	Supervisor	Assign or reassign a ticket to an eligible active agent within scope.
UC-004	Update Ticket Status	Agent, Supervisor	Move a ticket through valid workflow statuses.
UC-005	Add Internal Comment	Agent, Supervisor, Manager	Add internal collaboration notes to a ticket.
UC-006	Escalate Ticket	Agent, Supervisor	Escalate a ticket requiring higher-level attention.
UC-007	Resolve Ticket	Agent, Supervisor	Mark a ticket as resolved and preserve resolution outcome.
UC-008	Close Ticket	Agent, Supervisor	Close a resolved ticket and complete the ticket lifecycle.
UC-009	View SLA Dashboard	Manager, Supervisor, Agent, Viewer	Review SLA and operational visibility within scope.
UC-010	Manage Users	Admin	Create, update, deactivate, assign roles, and assign scopes to users.

11.1 Core Use Case Acceptance Pattern

Each use case should define actor, goal, preconditions, main flow, alternative flows, business rules, postconditions, and acceptance criteria. Use cases must validate that OpsSphere is not only a collection of screens or database tables, but a structured operational platform that supports real business execution.

12. Data and Domain Requirements

12.1 Core Domain Concepts

Region, Country, Account, Campaign, User, Role, Permission, Scope, Manager, Supervisor, Agent, Viewer, Customer, Ticket, Ticket Status, Ticket Priority, SLA, SLA State, Escalation, Queue, Comment, Resolution, Audit Trail, Dashboard, Reporting View.

12.2 Ticket Context

A ticket is the central operational entity. Each ticket should preserve enough context to answer who owns it, where it belongs, how urgent it is, what SLA condition it is in, what changes occurred, and who is accountable for oversight.

- Customer
- Region
- Country
- Account
- Campaign
- Assigned Agent
- Supervisor
- Priority
- Status
- SLA
- Comments
- Escalations
- Resolution
- Audit Trail

12.3 Role, Permission, and Scope Model

OpsSphere authorization is based on three dimensions:

Role + Permission + Operational Scope

A user is authorized only when the user is authenticated, has the required role or permission, and the target record belongs to the user assigned operational scope.

12.4 Audit and Historical Traceability

Historical traceability is a core requirement. User, organizational, and ticket records that are referenced by historical tickets should normally be deactivated rather than physically deleted. Ticket history, assignment history, escalation history, resolution data, and audit logs must preserve the business context of operational events.

12.5 Reporting-Ready Data Expectations

OpsSphere should maintain consistent operational fields that allow dashboards, filters, CSV exports, and future Power BI integration. The MVP should focus on clean structured data rather than advanced analytics.

13. Assumptions and Constraints

13.1 Business Assumptions

- The organization operates as a multinational BPO/contact center.
- The business structure includes regions, countries, accounts, campaigns, managers, supervisors, agents, viewers, roles, and permissions.
- Administrators maintain the platform configuration.
- Managers oversee one or more regions.
- Supervisors oversee accounts and campaigns within assigned scope.
- Agents work within assigned accounts or campaigns.
- Viewers require read-only access to operational information.
- Customers do not directly access the initial system.
- Advanced reporting is handled outside OpsSphere by tools such as Power BI.

13.2 Technical Assumptions

- Backend uses .NET 8 and ASP.NET Core Web API.
- Frontend uses Angular and TypeScript.
- SQL Server is the primary relational database.
- Entity Framework Core is the primary data access technology.
- Clean Architecture and CQRS with MediatR guide backend structure.
- JWT authentication, role-based authorization, and scope-based access are enforced by backend services.
- Docker, GitHub Actions, and Azure-ready patterns support delivery planning.

13.3 Scope Constraints

- The first version must focus on operational management and avoid becoming too broad.
- Advanced BI, telephony, workforce management, AI automation, and omnichannel integrations are excluded from the MVP.
- The MVP should prioritize ticket lifecycle, role-based access, SLA tracking, audit history, and basic dashboards.
- All sample accounts, campaigns, users, tickets, and operational examples must be fictional.

13.4 Security Constraints

- All protected features require authentication.
- Backend authorization must enforce role, permission, and scope.
- Sensitive secrets, tokens, passwords, and internal stack traces must not be exposed to users.
- Customers must not authenticate into the MVP.
- Frontend visibility rules must not replace backend authorization enforcement.

14. Acceptance Criteria

14.1 Business Acceptance Criteria

- OpsSphere represents a realistic multinational BPO/contact center operational structure.
- The platform supports operational visibility across regions, countries, accounts, campaigns, supervisors, and agents.
- The platform improves visibility into ticket ownership, status, priority, SLA state, and escalation history.
- The platform produces structured operational data suitable for external reporting tools.

14.2 Functional Acceptance Criteria

- Users can authenticate into the system.
- Users can access only features allowed by their role and operational scope.
- Admins can configure operational structure, users, roles, permissions, and scopes.
- Tickets can be created, assigned, updated, escalated, resolved, and closed.
- Tickets can be filtered by region, country, account, campaign, supervisor, agent, status, priority, and SLA state.
- SLA state can be tracked for tickets.
- Internal comments can be added to tickets by authorized users.
- Important ticket and administrative changes are recorded in audit history.
- Basic dashboards provide operational visibility.

14.3 Technical Acceptance Criteria

- The backend follows Clean Architecture principles.
- Business logic is not implemented directly inside controllers.
- The system uses JWT authentication and backend-enforced role/scope authorization.
- The system persists operational data in SQL Server.
- The system includes automated tests for core workflows and access rules.
- The local development environment can be containerized with Docker.
- The project includes CI/CD-ready structure using GitHub Actions.
- The solution is designed with Azure-ready deployment practices.
- The system exposes GET /health and GET /health/details as operational health endpoints.

14.4 Portfolio Acceptance Criteria

- The document is professional, readable, and consistent with the Markdown documentation.
- The SRS can be used as a portfolio artifact and implementation reference.
- The SRS demonstrates business analysis, requirements engineering, architecture thinking, security awareness, and delivery planning.
- The formal PDF generated from this editable document can be placed under docs/formal-specification/.

15. Success Criteria

15.1 Business Success Criteria

- Administrators can manage regions, countries, accounts, campaigns, managers, supervisors, agents, viewers, roles, and permissions.
- Managers can review operational performance across assigned regions.
- Supervisors can monitor workload, SLA status, escalations, and agents within assigned scope.
- Agents can manage tickets within assigned account or campaign.
- Viewers can access read-only operational information.
- The system supports external reporting by producing structured operational data.

15.2 Functional Success Criteria

- Authentication, role-based access, operational scope, ticket lifecycle, SLA tracking, comments, escalations, audit history, dashboards, and filtered views work together as a cohesive MVP.
- Critical negative cases are handled: invalid login, deactivated user, missing role, unauthorized scope, invalid status transition, closed ticket modification, missing escalation reason, and viewer write attempt.

15.3 Technical Success Criteria

- The solution demonstrates senior-level .NET backend practices, Angular frontend planning, SQL Server database design, authentication and authorization, testing, Docker, CI/CD, and Azure-ready deployment.
- The architecture remains modular enough to add Phase 2 and Phase 3 capabilities without rewriting the core domain.

16. Glossary

Term	Definition
Region	Large geographic operating area such as Latin America, North America, EMEA, or APAC.
Country	National operation within a region.
Account	Client account served by the BPO/contact center.
Campaign	Operational service or workload inside an account.
Manager	User responsible for overseeing one or more regions.
Supervisor	User responsible for overseeing agents and tickets within an account or campaign.
Agent	User responsible for handling tickets.
Viewer	Read-only user who accesses operational information.
Customer	Person or entity associated with a support request.
Ticket	Operational support case handled by agents and supervisors.
SLA	Service Level Agreement target associated with a ticket.
SLA State	Current SLA condition such as Within SLA, At Risk, Breached, or Completed.
Escalation	Action used when a ticket requires supervisor or higher-level attention.
Audit History	Record of important changes made in the system.
Operational Dashboard	Visibility screen for ticket, SLA, workload, and operational status.
RBAC	Role-Based Access Control.
Scope	Operational visibility boundary assigned to a user.
Power BI	External business intelligence tool used for advanced reporting outside the core application.
Health Check	Operational endpoint that reports application and dependency availability.

17. Appendices

Appendix A: Source Document Mapping

SRS Section	Primary Source Documents
Introduction	00-executive-summary.md, 03-project-charter.md, 06-requirements.md
Business Context	01-business-context.md, 02-business-case.md, 03-project-charter.md
Project Scope	03-project-charter.md, 05-scope-and-roadmap.md, 06-requirements.md
Stakeholders	04-stakeholders.md, 06-requirements.md, 07-use-cases.md
Functional Requirements	06-requirements.md, 07-use-cases.md, 09-business-rules.md
Non-Functional Requirements	06-requirements.md, 11-architecture-overview.md, 16-security-and-permissions.md, 17-testing-strategy.md, 18-deployment-and-devops.md, 19-observability-and-support.md
Business Rules	09-business-rules.md
Use Cases	07-use-cases.md
Domain and Data Requirements	10-domain-model.md, 14-database-design.md
Assumptions, Constraints, Success Criteria	03-project-charter.md, 05-scope-and-roadmap.md

Appendix B: Requirement Category Summary

Category	Requirement Count/Scope
Authentication and Authorization	Login, JWT, protected resources, RBAC, scope visibility, authorization errors, session expiration.
User and Role Management	User lifecycle, role assignment, scope assignment, deactivation, user audit history.
Operational Structure	Regions, countries, accounts, campaigns, assignments, deactivation, structure audit history.
Ticket Lifecycle	Creation, assignment, status workflow, comments, escalation, resolution, closure, ticket history.
SLA	SLA target, state, at-risk, breached, completed, dashboard visibility.
Audit	Actor, timestamp, entity, action, previous/new values, immutability, scoped visibility.
Reporting	Dashboard metrics, filtering, CSV export boundary, Power BI boundary.

Appendix C: Graphic Placeholders

- C4 System Context Diagram.
- Operational Hierarchy Diagram.
- UML Use Case Diagram.
- Domain Class Diagram.
- Ticket Lifecycle Process Diagram.
- SLA Monitoring Process Diagram.
- Escalation Process Diagram.

Appendix D: Health Endpoint Decision

The audit identified conflicting documentation between /health and /api/health. This SRS standardizes operational health checks as GET /health and GET /health/details. The API design documentation should be corrected if it still references /api/health as the default health endpoint.